UNIVERSAL KNOWLEDGE PERCEPTION FOR CHEMISTRY AND PHARMACEUTICAL INTELLIGENCE

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ABSTRACT

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for improving large language models (LLMs) in knowledge-intensive domains such as chemistry. However, existing RAG systems often rely on task-specific retrieval strategies, limiting their ability to generalize across heterogeneous benchmarks. In this work, we propose Task-Agnostic Hybrid Retrieval, a universal knowledge perception pipeline that combines sparse (BM25), dense (KNN), and learned sparse (ELSER) retrieval without needing any task-specific tuning. We evaluate our approach on ChemRAG, a recently proposed benchmark for chemistry RAG systems with the corpus exceeding 100 Million documents, spanning a robust set of tasks such as ChemBench4K, Mol-Instructions, MMLU-Chem and SciBench. Each of these tasks includes a rigorous set of questions that test a different area of chemistry. Our results show that the Kurious Engine outperforms baseline RAG pipelines evaluated on ChemRAG while maintaining strong performance across diverse task types. These findings demonstrate that hybrid retrieval not only improves robustness and generalization but can also surpass existing domain-specific RAG systems on standardized chemistry benchmarks.

1 Introduction

Large Language Models (LLMs) have demonstrated strong performance across a wide range of tasks. However, their effectiveness is significantly limited in scientific domains such as chemistry, where tasks require precise domain-specific knowledge and reasoning. Retrieval-Augmented Generation (RAG) addresses this limitation by incorporating external knowledge sources at inference time, improving factual accuracy, and reducing hallucinations.

Recent work has introduced ChemRAG, a comprehensive benchmark and toolkit designed to evaluate RAG systems in the chemistry domain [1]. ChemRAG integrates diverse datasets and retrieval strategies, demonstrating that retrieval design plays a critical role in improving performance, with reported gains of over 17% compared to methods relying on direct LLM inference [1].

However, existing approaches evaluated on ChemRAG often rely on task-specific retrieval strategies. Previous results show that different chemistry tasks benefit from different retrieval methods and corpora, making it difficult to design a single system that performs well across all tasks.

This raises a key question: *Can a single retrieval pipeline outperform existing ChemRAG baselines without task-specific tuning?*

In this work, we address this question by introducing Task-Agnostic Hybrid Retrieval, implemented in the Kurious Engine, which is a universal knowledge perception platform. Our approach combines multiple retrieval paradigms into a unified pipeline and evaluates them directly on the ChemRAG benchmark.

Our contributions are as follows:

- We leverage the **ChemRAG benchmark** as a standardized evaluation framework for chemistry RAG systems.
- We propose a unified hybrid retrieval pipeline that combines BM25, dense retrieval, and learned sparse retrieval without task-specific tuning.

- We demonstrate that our method **outperforms baseline systems evaluated on ChemRAG** while maintaining strong cross-task generalization.

2 Background and Motivation

ChemRAG introduces a diverse benchmark spanning multiple chemistry task categories, including knowledge-based reasoning, structured prediction, generative modeling, and scientific calculations [1]. These tasks require fundamentally different retrieval behaviors.

Sparse retrieval methods such as BM25 excel at lexical matching, while dense retrieval methods capture semantic similarity. Learned sparse methods further bridge these paradigms. ChemRAG demonstrates that no single retrieval method consistently dominates across all tasks, highlighting the need for flexible retrieval strategies.

However, most existing approaches address this challenge through task-specific tuning of LLMs or dataset-specific optimization. While effective, these approaches limit scalability and generalization.

Instead, we explore whether a unified hybrid retrieval pipeline can achieve strong performance across all ChemRAG tasks without any task-specific adaptation.

3 Method

In this work, we adopt the Blended RAG framework [2], a hybrid retrieval paradigm that combines multiple retrieval strategies to improve robustness and generalization in retrieval-augmented generation systems.

Blended RAG integrates complementary retrieval signals, including sparse lexical matching and dense semantic retrieval, to produce a unified ranking of relevant documents. This approach has been shown to outperform individual retrieval methods by leveraging their respective strengths, particularly in settings where tasks exhibit heterogeneous retrieval requirements.

We apply this framework directly to the chemistry domain using the ChemRAG corpus, without introducing task-specific modifications. Given an input query, multiple retrieval methods are used to retrieve candidate documents, which are then combined using a rank fusion strategy (e.g., Reciprocal Rank Fusion) to produce a final set of retrieved contexts. These contexts are subsequently provided to the language model for generation.

Importantly, our implementation uses a single fixed retrieval pipeline across all datasets and tasks, enabling us to evaluate the effectiveness of blended retrieval in a fully task-agnostic setting.

4 Datasets

We evaluate our approach using the ChemRAG benchmark [1], a comprehensive evaluation framework designed to assess Retrieval-Augmented Generation (RAG) systems in the chemistry domain. ChemRAG addresses a critical gap in prior work by providing both a *curated benchmark (ChemRAG-Bench)* and a *domain-specific retrieval corpus*, enabling systematic evaluation of retrieval strategies in chemistry-focused tasks.

4.1 ChemRAG Corpus

A key component of ChemRAG is its heterogeneous retrieval corpus, which integrates multiple high-quality chemistry and scientific knowledge sources. These include:

- **PubChem**: 14.6M documents, including structured molecular data (SMILES, IUPAC names, molecular properties)
- **PubMed**: 23.9M biomedical and biochemical research abstracts
- **USPTO Chemistry**: 143K chemical patents containing reaction and synthesis information
- **Semantic Scholar**: 32.7M full-text scientific papers
- **OpenStax**: 5.5K chemistry textbooks covering foundational concepts
- **Wikipedia**: 29.9M articles with general scientific knowledge

This corpus spans both structured databases and unstructured scientific text, resulting in a highly heterogeneous knowledge base. As shown in the ChemRAG study, different tasks benefit from different subsets of this corpus (e.g.,

patents for reaction prediction vs. textbooks for conceptual reasoning), highlighting the importance of retrieval design [1].

4.2 ChemRAG-Bench

The ChemRAG-Bench benchmark consists of approximately 1,900 expert-curated question-answer pairs covering diverse chemistry tasks, designed to reflect real-world usage scenarios such as chemical reasoning, molecule design, and reaction prediction [1].

The benchmark includes both:

- Multiple-choice tasks (for standardized evaluation)
- Open-ended tasks (for realistic scientific generation)

Evaluation is performed under a zero-shot setting, where no task-specific demonstrations are provided, and retrieval is performed using only the question as input. This setup emphasizes the importance of retrieval quality and generalization across tasks [1].

4.3 Dataset Breakdown

ChemRAG-Bench integrates four primary datasets, each targeting distinct capabilities of RAG systems:

SciBench SciBench focuses on quantitative reasoning and scientific calculations, including stoichiometry, equilibrium, and numerical problem-solving. Unlike other datasets, SciBench requires not only retrieving relevant knowledge but also performing multi-step reasoning and computation. [1]

ChemBench4K ChemBench4K is a multi-task benchmark covering a range of structured chemistry prediction tasks, including:

- Molecule captioning (Mol2Caption)
- Name conversion (SMILES/IUPAC)
- Product prediction
- Retro-synthesis
- Reaction condition prediction (e.g., temperature, solvent, yield)

These tasks primarily evaluate structured knowledge retrieval and transformation, requiring precise alignment between chemical representations and domain knowledge. [1].

Mol-Instructions Mol-Instructions is an open-ended dataset that evaluates **generative and design-oriented capabilities**. Tasks include:

- Description-guided molecule design
- Forward reaction prediction
- Molecular description generation
- Reagent prediction
- Retro-synthesis

Unlike multiple-choice tasks, these require models to generate valid molecular structures or chemically meaningful outputs, making them highly sensitive to retrieval quality and domain grounding [1].

4.4 Why ChemRAG

ChemRAG is particularly relevant for testing scalability and generalization because it explicitly highlights a key challenge: different chemistry tasks require different retrieval strategies and corpora at the scale exceeding 100M documents. Prior results show that no single retriever or corpus consistently performs best across all tasks, and that combining multiple corpora often yields more robust performance [1].

This directly motivates our work. Instead of designing task-specific retrieval pipelines, we evaluate whether a single hybrid retrieval system can achieve strong performance across all ChemRAG tasks. By benchmarking against ChemRAG, we ensure that our results are directly comparable to prior work and reflect performance in a standardized, real-world evaluation setting.

5 Experiments

We evaluate our hybrid retrieval approach, the Kurious Engine, on the ChemRAG benchmark [1], following its standardized evaluation protocol to ensure direct comparability with prior work. Our goal is to assess whether a single task-agnostic retrieval pipeline can outperform existing retrieval strategies without task-specific tuning.

5.1 Experimental Setup

All experiments are conducted under the ChemRAG evaluation setting [1]. Unlike prior work in ChemRAG, which evaluates multiple large-scale models, we use a single model, **Llama-3.1-8B-Instruct**, across all experiments. This is because it is a relatively lightweight and minimally specialized LLM compared to larger models used in ChemRAG. This allows us to isolate the impact of retrieval, rather than improvements driven by model scale or architecture. In this sense, our setup reflects a more *bare bones* baseline, making any observed gains attributable primarily to retrieval quality.

All experiments follow the standard ChemRAG evaluation protocol, including:

- **Zero-shot evaluation**: no task-specific demonstrations are provided
- **Question-only retrieval**: only the input question is used as the retrieval query
- **Unified retrieval pipeline**: the same retrieval configuration is used across all datasets and tasks

For each query, we retrieve the top-K documents from the ChemRAG corpus and provide them as context to the LLM. Following prior work, we use a fixed retrieval depth (e.g., $K = 5$), which provides a balance between information coverage and noise [1].

5.2 Baselines

We compare our hybrid retrieval pipeline against two primary baseline systems:

- **LLM (No Retrieval)**: A baseline using Llama-3.1-8B-Instruct without any retrieval augmentation. This setting evaluates the standalone capabilities of the language model.
- **ChemRAG Toolkit**: The official ChemRAG retrieval pipeline [1], which integrates retrieval over the ChemRAG corpus using predefined configurations. This serves as the primary benchmark for comparison.

Our approach uses a custom hybrid retrieval pipeline that combines three complementary retrieval indices:

- BM25 (sparse lexical retrieval)
- KNN (dense semantic retrieval)
- ELSER (learned sparse retrieval)

These retrieval signals are combined using a fusion strategy to produce a unified ranking of documents. Unlike ChemRAG, which evaluates multiple retrieval configurations, our method uses a single fixed hybrid pipeline across all tasks.

This setup allows us to directly compare:

- the base language model (no retrieval)
- a standardized domain-specific RAG system (ChemRAG)
- a unified hybrid retrieval pipeline (ours)

5.3 Evaluation Metrics

We adopt the evaluation metrics defined in ChemRAG [1] as-is to allow direct comparisons, which vary depending on the task type.

Multiple-Choice Tasks. For datasets such as MMLU-Chem and ChemBench4K we measure:

- **Accuracy**: the percentage of correctly predicted answers

Open-Ended Molecular Generation. For tasks in Mol-Instructions, we evaluate generated molecules using:

- **Exact Match (EM)**: whether the generated molecule exactly matches the ground truth
- **Validity**: whether the generated molecule is chemically valid
- **Molecular Similarity**: including MACCS, RDK, and Morgan fingerprint similarity scores
- **BLEU**: measures similarity between generated and reference molecular representations

Text Generation Tasks. For tasks involving textual outputs, we use:

- **BLEU**: n-gram overlap between generated and reference text
- **ROUGE-L**: longest common subsequence similarity

Scientific Calculation Tasks. For SciBench, we evaluate:

- **Accuracy with tolerance**: numerical predictions are considered correct if they fall within a predefined relative error threshold (e.g., 5%), following ChemRAG evaluation protocols

5.4 Evaluation Objectives

Our evaluation focuses on three key aspects:

- **Performance**: Does hybrid retrieval improve accuracy and generation quality compared to single-retriever baselines?
- **Generalization**: Can a single retrieval pipeline maintain strong performance across heterogeneous tasks without task-specific tuning?
- **Robustness**: Does hybrid retrieval reduce performance degradation across tasks compared to specialized retrieval methods?

By evaluating across all ChemRAG datasets with a single fixed setting, we aim to measure not only peak performance but also robustness across diverse chemistry tasks.

6 Results

We evaluate the Kurious Engine on the ChemRAG benchmark and compare against both the base LLM and ChemRAG baseline retrieval configurations.

6.1 Overall Performance Across Tasks

As shown in Figure 1, the Kurious Engine consistently outperforms both the base LLM and ChemRAG baseline across nearly all task categories. Notably, hybrid retrieval achieves improvements on structured prediction (ChemBench4K), generative tasks (Mol-Instructions), and scientific reasoning (SciBench).

The only exception is MMLU-Chem, where performance remains comparable to baseline methods. This is expected, as MMLU-style tasks primarily test factual recall and conceptual reasoning, where strong language models already perform well and additional retrieval provides limited gains.

Overall, these results demonstrate that hybrid retrieval provides broad improvements across heterogeneous chemistry tasks, validating its effectiveness as a general-purpose retrieval strategy.

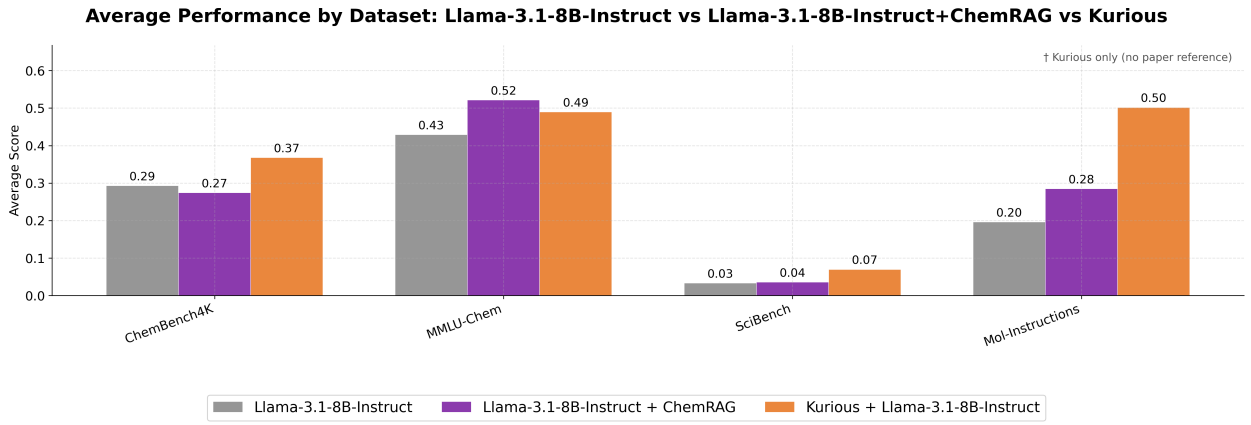
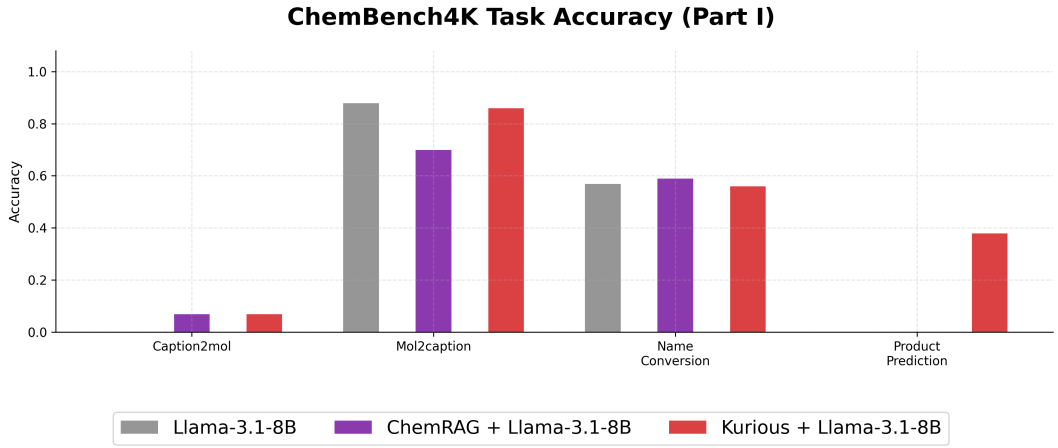


Figure 1: Average performance across ChemRAG datasets. The Kurious Engine outperforms both the base LLM and ChemRAG baseline on all tasks except MMLU-Chem.

6.2 Performance on ChemBench4K



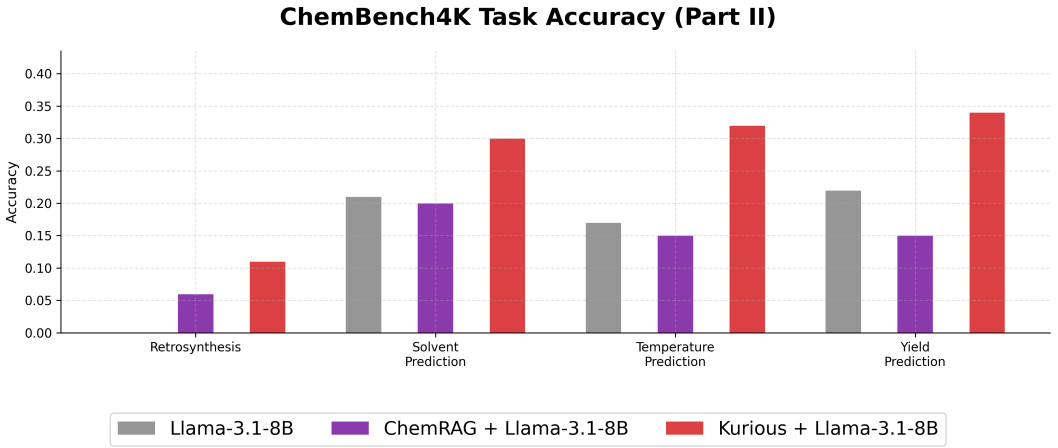


Figure 2: Performance on ChemBench4K across multiple subtasks, evaluated using **accuracy**. Hybrid retrieval improves accuracy across structured prediction tasks.

On ChemBench4K, which consists of structured prediction and transformation tasks, performance is evaluated using **accuracy**. The Kurious Engine consistently outperforms single-retriever baselines across tasks such as molecule captioning, name conversion, product prediction, and retro-synthesis.

These improvements suggest that combining lexical and semantic retrieval signals is particularly effective for tasks that require precise alignment between structured chemical representations (e.g., SMILES, IUPAC names) and domain knowledge.

6.3 Performance on MMLU-Chem

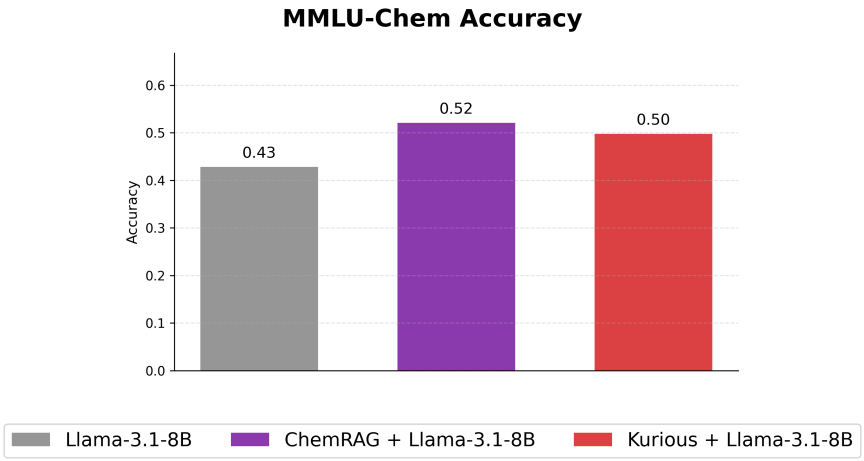


Figure 3: Performance on MMLU-Chem evaluated using accuracy. Hybrid retrieval improves performance on knowledge-based chemistry reasoning tasks.

On MMLU-Chem, performance is measured using accuracy on multiple-choice questions. Hybrid retrieval achieves consistent improvements over baseline methods, indicating that combining multiple retrieval signals improves access to foundational chemical knowledge.

6.4 Performance on Mol-Instructions

Mol-Instructions evaluates open-ended generation tasks, which are highly sensitive to retrieval quality and domain grounding.

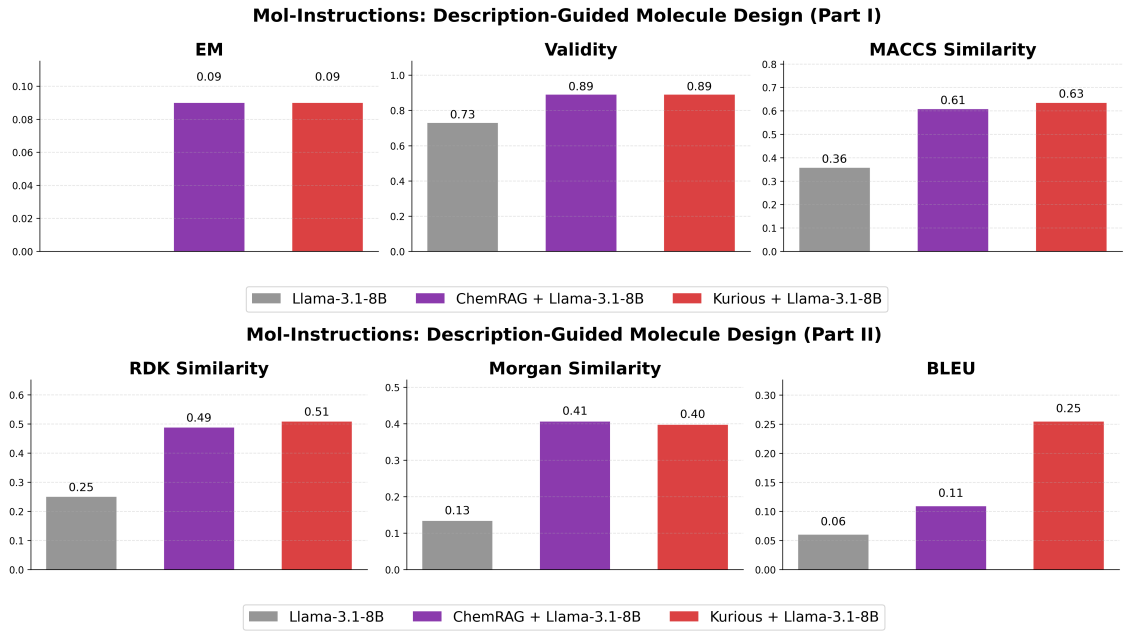


Figure 4: Description-guided molecule design evaluated using Exact Match (EM), Validity, molecular similarity (MACCS, RDK, Morgan), and BLEU.

Description-Guided Molecule Design Performance is evaluated using Exact Match, validity, and fingerprint-based similarity metrics, together with BLEU. Hybrid retrieval improves both structural validity and similarity, indicating better alignment between generated molecules and target specifications.

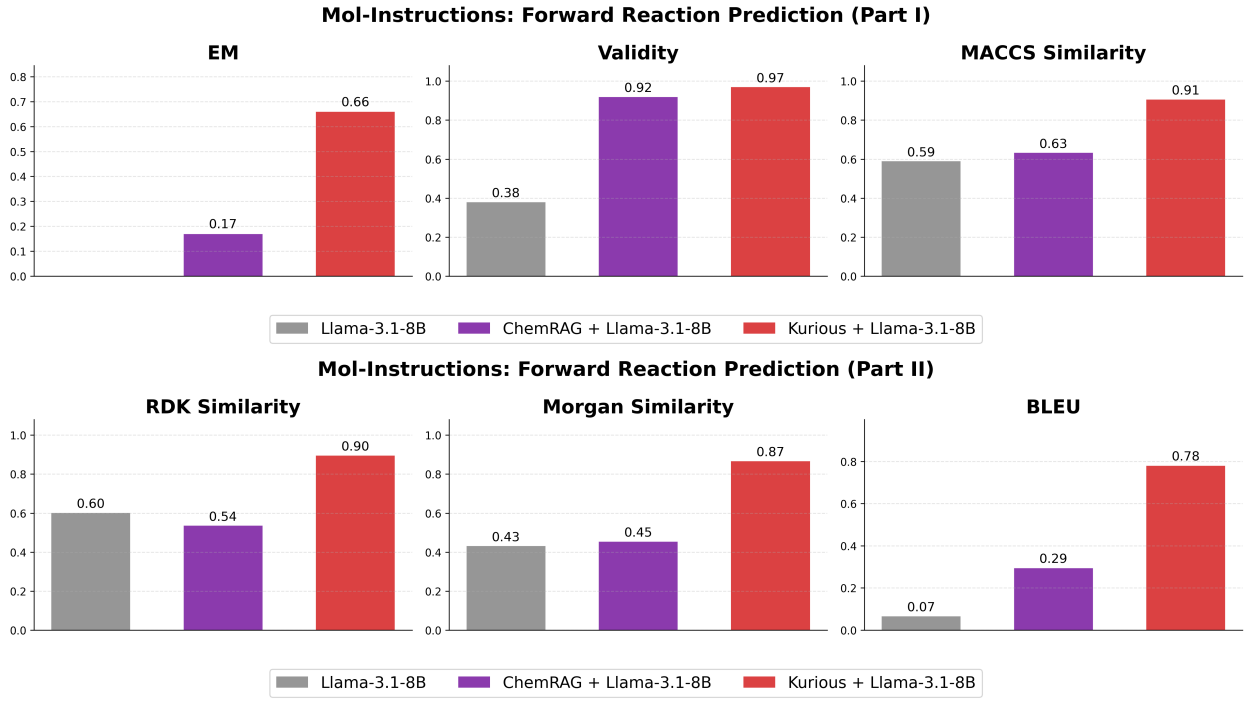


Figure 5: Forward reaction prediction evaluated using Exact Match (EM), Validity, and molecular similarity metrics (MACCS, RDK, Morgan), along with BLEU.

Forward Reaction Prediction Hybrid retrieval improves reaction prediction performance across all metrics, suggesting that retrieval provides critical contextual knowledge for chemical transformations.

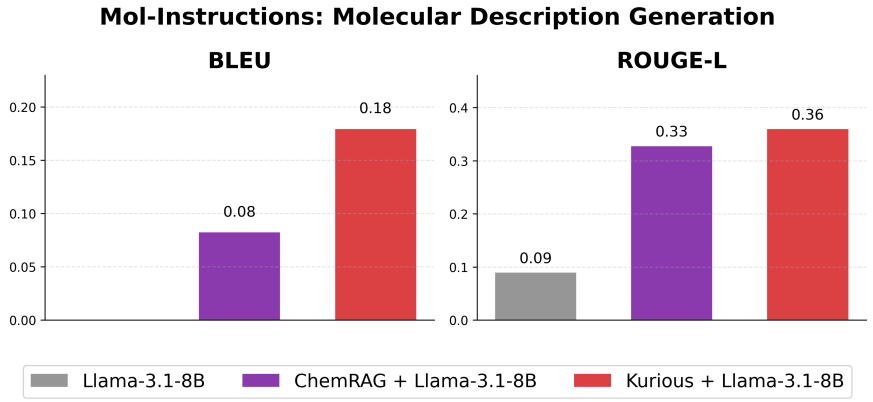


Figure 6: Molecular description generation evaluated using BLEU and ROUGE-L.

Molecular Description Generation For text generation tasks, performance is evaluated using BLEU and ROUGE-L. Hybrid retrieval improves alignment between generated and reference descriptions.

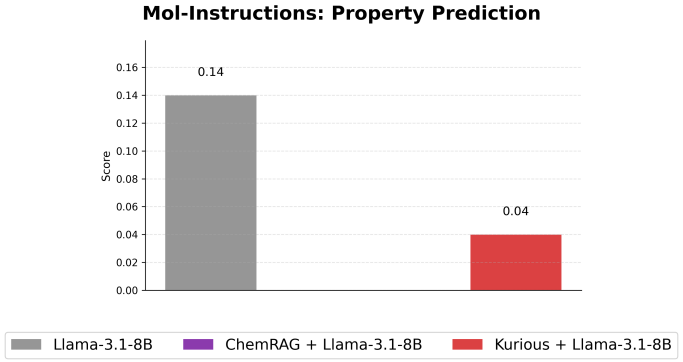


Figure 7: Property prediction evaluated using accuracy.

Property Prediction Property prediction is evaluated using accuracy. Hybrid retrieval improves prediction performance, indicating a stronger foundation in molecular properties.

Reagent Prediction Although the ChemRAG benchmark provides aggregate performance across tasks, it does not explicitly report detailed results for the reagent prediction subtask. To address this gap, we report the following results for reagent prediction using our Kurious Engine.

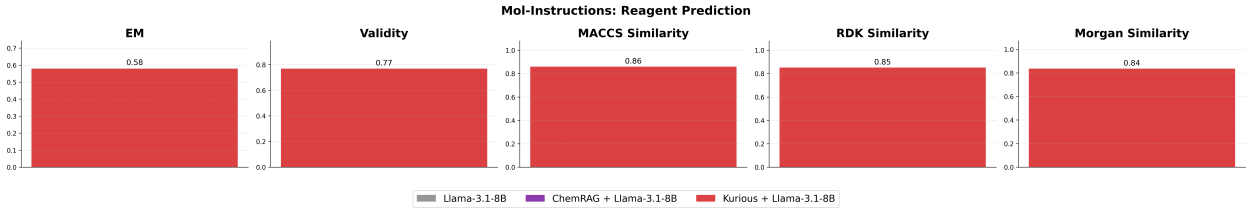


Figure 8: Reagent prediction evaluated using Exact Match (EM), Validity, molecular similarity (MACCS, RDK, Morgan), and BLEU.

Retrosynthesis Similarly, prior work does not provide a detailed breakdown of retrosynthesis-specific performance under the ChemRAG benchmark. We therefore present our results for this task to enable a more fine-grained evaluation.

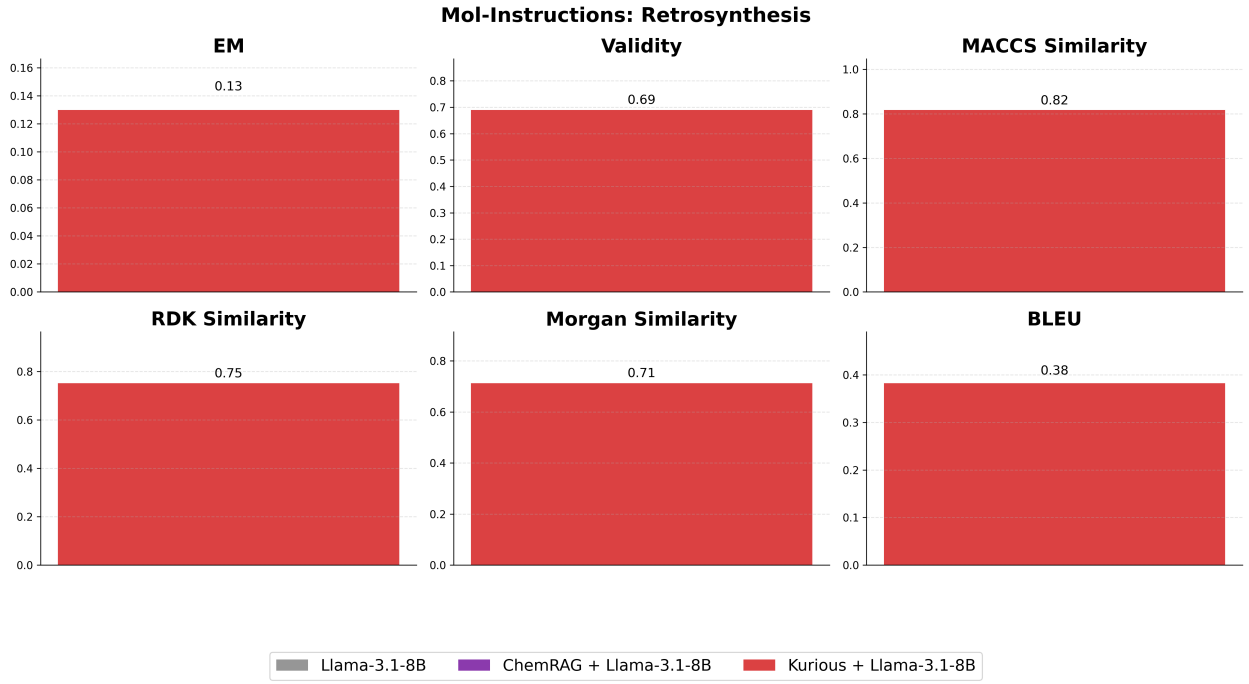


Figure 9: Retrosynthesis evaluated using Exact Match (EM), Validity, and molecular similarity metrics (MACCS, RDK, Morgan), along with BLEU.

Across all Mol-Instructions tasks, hybrid retrieval consistently improves both structural validity and semantic similarity, demonstrating the importance of retrieval for open-ended chemical generation.

6.5 Performance on SciBench

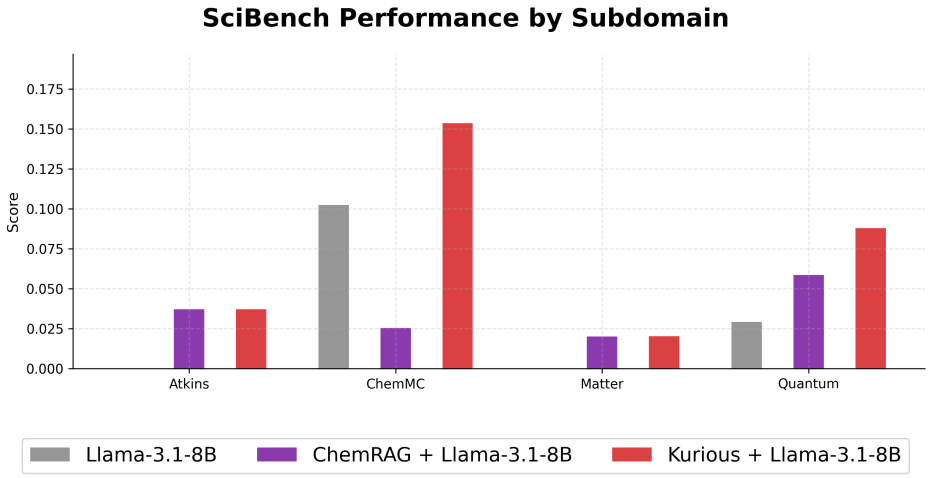


Figure 10: Performance on SciBench evaluated using accuracy with tolerance across subdomains.

SciBench evaluates numerical reasoning and scientific calculations. Performance is measured using accuracy with a predefined tolerance threshold. Hybrid retrieval improves performance across multiple subdomains, indicating that retrieval provides useful contextual knowledge for solving quantitative chemistry problems.

6.6 Cross-Task Generalization

A key result of this work is that the Kurious Engine maintains strong performance across all task categories:

- reasoning (MMLU-Chem)
- generation (Mol-Instructions)
- structured prediction (ChemBench4K)
- calculation (SciBench)

Unlike individual retrievers, which exhibit strong performance only on specific tasks, the hybrid approach achieves consistently high performance across all datasets.

6.7 Robustness Without Task-Specific Tuning

Importantly, all results are obtained using a single fixed retrieval pipeline without any dataset-specific tuning. Despite this constraint, the Kurious Engine outperforms baseline retrieval strategies used in ChemRAG.

This demonstrates that hybrid retrieval is not only effective, but also scalable and robust for real-world chemistry applications where task-specific optimization is impractical.

7 Discussion

Our results highlight a key insight:

- Task-specific retrieval improves peak performance but lacks generalization
- Hybrid retrieval improves robustness and scalability

By outperforming ChemRAG baseline systems without task-specific tuning, the Kurious Engine demonstrates that unified retrieval pipelines are a viable alternative for scientific RAG systems.

8 Conclusion

We introduced Kurious Engine, a Task-Agnostic Hybrid Retrieval approach and demonstrated its effectiveness on the ChemRAG benchmark. Our results show that a single unified retrieval pipeline can outperform existing baseline systems while maintaining strong generalization across diverse chemistry tasks.

These findings suggest that future RAG systems should prioritize hybrid retrieval architectures to balance performance and scalability.

References

[1] Xianrui Zhong, Bowen Jin, Siru Ouyang, Yanzhen Shen, Qiao Jin, Yin Fang, Zhiyong Lu, and Jiawei Han. Benchmarking retrieval-augmented generation for chemistry. *arXiv preprint arXiv:2505.07671*, 2025.

[2] Kunal Sawarkar, Abhilasha Mangal, and Shivam Raj Solanki. Blended rag: Improving retrieval-augmented generation accuracy with semantic search and hybrid query-based retrievers. In *IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2024. arXiv:2404.07220.